

Kuan-Cheng (James) Ku

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Education

- National Taiwan University** *Master in Information Management* Feb 2026 – Present
- National Taiwan University** *Bachelor in Information Management (3.88 GPA)* Sep 2021 – June 2025
- Key courses include Data Structures, Algorithms, Database, Web Programming, Deep Learning, Information Retrieval & Text Mining, Cloud Native Application, Systems Analysis & Design, Management of Hi-tech Company
 - Teaching assistant for 1000+ students across 7 courses, including Web Programming (EE), Web APP Development (IM), Machine Learning, Data Structures, Systems Analysis & Design, Calculus, and Marketing Management (EMI course).

Work Experience

- Research Development Intern, Microsoft Taiwan** *TSMC OnePlatform Team, ISD* Apr 2025 – Jan 2026
- Led e2e testing from design to acceptance, executing 400+ test cases and coordinating QA-client validation.
 - Identified 150+ bugs, built frontend analysis POC, and led cross-vendor meetings to accelerate bug resolution.
 - Authored 300+ page LaTeX manual and delivered critical training sessions instrumental to successful project closure.
- Contract Developer, National Chung Hsing University** *NCHU Buddy Program* Apr 2025 – Sep 2025
- Architected and developed a full-stack student management dashboard using Next.js, Supabase, and Google Cloud API.
 - Built task management, dynamic forms, real-time analytics, notifications, and Google Drive integration with RBAC.
 - Shipped production platform serving 300+ MAU, completed client acceptance, and successfully concluded project handoff.

Research Experience

- Research Assistant, National Taiwan University Hospital** *Peritoneal Dialysis Center* Jan 2026 – Present
- Building a smart healthcare app for PD patients to assess exit-site infection risk at home using YOLO and CNN models.
 - Implemented patient-facing exit-site photo capture and a nurse-facing backend to view image history and AI results.
- SARS: Social Advertising Recommendation System** 📄 [Report](#) 📄 [Slides](#) Nov 2023 – Jan 2024
- Developed ad engine integrating BERT-based comment scoring and Dijkstra-based user subgraph modeling.
 - Optimized retrieval parameters using Gurobi solver and systematic backtesting.
- StreamBench Agent: Continuous Improvement of LLM** 📄 [Presentation](#) 📄 [Report](#) Oct 2024 – Dec 2024
- Built StreamBench-inspired pipeline for medical diagnosis LLM, improving accuracy from 50.17% to 73.64%.
 - Engineered prompting techniques, including clinical roleplay, structured reasoning (CoT) and negative example learning.

Achievements

- IEC 2025, Honorable Mention: RenderResume** 📄 [Product](#) 📄 [Demo](#) 🏆 [Certificate](#) Jun 2025 - Oct 2025
- Built RenderResume, a resume platform featuring AI analysis and chat agent with in-context editing capabilities.
 - Gained 300+ users & 3,000+ uses during Beta, validating product-market fit as strategic entry into job matching market.
 - Selected as Top 8 finalist from 200+ teams nationally.
- TSMC IT CareerHack 2024, 2nd Place: deVopSCode** 📄 [GitHub](#) 🏆 [Certificate](#) Feb 2025
- Developed an AI maintenance toolkit with web and VSCode interfaces using Vertex AI, FastAPI and LangChain
 - Rapidly prototyped and deployed the full-stack solution with GitHub integration on GCE and GKE within 2 days.
 - Selected as one of 24 finalists from 168 teams nationally and placed second.

Projects

- One-Click Outfit** 📄 [GitHub](#) 📄 [Demo](#) 📄 [Report](#) 📄 [Poster](#) 📄 [Slides](#) Jan 2024 – Jan 2025
- Led team of 4 to build AI fashion recommendation platform using Next.js and Flask for personalized styling.
 - Improved performance 2.4x by migrating to PostgreSQL with vector indexing and Docker deployment on VPS.
- Customized IT Solution for ShiJiou Cafe** 📄 [Report](#) 📄 [Slides](#) Mar 2023 – May 2023
- Developed automated customer flow system for local coffee shop, addressing wait times and queue inefficiencies.
 - Built online menu, LINE chatbot queueing, and unified backend using React, Express, Firebase & LINE Messaging API.

Volunteering and Leadership

Lecturer, Taipei Municipal Datong High School <i>Information Research Club</i>	Sep 2022 – Jan 2026
Lecturer, Taipei Municipal Zhongshan Girls High School <i>Information Research Club</i>	Feb 2024 – Jan 2026
Lecturer, NTU CSIE Sprout Program ▶ Example Lecture 📄 Example Slides	Dec 2021 – Jun 2024
◦ Designed and delivered C++ programming courses to foster pre-univ students’ interest in computer science, build foundations for APCS preparation, and guide career exploration in CS fields.	
Head of IT, NTU Information Management Student Association	Jun 2022 – Jan 2024
◦ Led development of a faculty interview platform to support students’ bachelor project advisor selection.	
◦ Co-administered Linux workstation systems and Google Workspace for the department.	
Co-founder, NTUIM Competitive Programming Team @ Instagram	Jan 2022 – Jun 2023
◦ Co-founded the team, recruited 2 lecturers and 36 members, and raised NT\$100K worth of rental space sponsorship.	
◦ Co-developed materials and delivered sessions on data structures and algorithms for competitive programming.	
◦ Regularly organized in-person, team-based ICPC-format contests on Codeforces for hands-on practice.	
Chief Organizer, 2023 NTUIM Programming Contest: PDAO f Facebook 📄 Article	Dec 2022 – May 2023
◦ Led organization of a nation-wide ICPC-format programming competition with cash flow exceeding NT\$100K.	
◦ Innovatively opened and coordinated synchronization of two off- and one on-campus venues to provide better experience.	
◦ Successfully recruited 81 teams comprising 226 participants, including 10+ professional competitive programming teams.	
◦ Achieved a post-competition satisfaction rating of 4.21/5, demonstrating strong participant engagement.	
Deputy Organizer, 2024 NTUIM Camp @ Instagram ▶ Youtube 📄 Brochure	Jun 2023 – Jan 2024
◦ Revived a 5-day camp from 4-year pandemic suspension by recruiting 60+ staff with cash flow exceeding NT\$800K.	
◦ Oversaw marketing and design, a cross-functional department of 10, comprising art, publicity and photography teams.	
◦ Led merchandise development, multi-channel event promotion, brand establishment, and comprehensive documentation, gaining 1K+ followers and 500K+ impressions on social media within 6 months, contributing to 200+ registrations.	
Dev Lead & Event Planner, NTU Art Festival 2024 <i>Riddle City Team</i> 🔗 Legacy	Jan 2024 – Jun 2024
◦ Led dev team of 3 to build treasure hunt game integrating web and physical campus riddles using Next.js and Supabase.	
◦ Bridged planning, design, and dev teams to launch game engaging 200+ users with 600+ gameplay hours.	
Staff Member, NTU Photography Club <i>Exhibitions Department</i> 🔗 Online Exhibition	Jul 2023 – Dec 2023
◦ Organized promotion booth and biannual exhibitions providing members with platforms to showcase their works.	
◦ Developed an online exhibition website using React.js and Firebase to expand exhibition reach and accessibility.	

Skills and Languages

Languages: Python, C/C++, HTML5, CSS3, JavaScript, TypeScript, SQL	Chinese: Native
Frameworks: Express, React, Next, Tailwind, LangChain, Flask, FastAPI, Pytorch	English: Fluent
Database: PostgreSQL, Supabase, Cloudflare D1 & R2, MongoDB, Firebase	TOEIC 950
Tools: Git, Docker, GCP, Vercel, Cloudflare, Azure DevOps, Playwright	
AI Tools: Cursor (1.6B+ tokens used in 2025 🔗) & Claude Code	